


```
Nyx says, "Great job, Ei! I've read that a lot of players create writeups of interesting challenges they solve during the competition. Just be sure to wait to publish them until after the winners have been announced. We can work on that together if you'd like."
```

```
----  
(Press Enter to continue...)  
----
```

```
"Thanks, Nyx! Here's the flag I found: picoCTF{m1113n1um_3d1710n_8dcbda00}"
```

```
----  
(Press Enter to continue...)  
----
```

Flag: picoCTF{m1113n1um_3d1710n_8dcbda00}

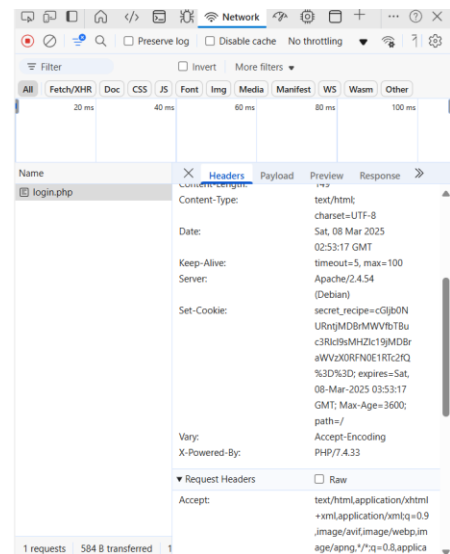
2 COOKIE MONSTER SECRET RECIPE

Access Denied

Cookie Monster says: "Me no need password. Me just need cookies!"

Hint: Have you checked your cookies lately?

[Go back](#)



Inspect cookies to find flag.

Decode from Base 64 format to text.

Decode from Base64 format

Simply enter your data then push the decode button.

```
cGJjb0NURntjMDBrMWVfbTBuc3Rlcl9sMHZlc19jMDBraWVzX0RfN0E1RTc2fQ%3D%3D
```

i For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8 Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

☒ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

```
picoCTF{c00k1e_m0nster_l0ves_c00kies_DE7A5E76}
```

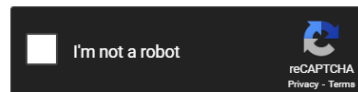
Flag: picoCTF{c00k1e_m0nster_l0ves_c00kies_DE7A5E76}

3 HASHCRACK

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

```
482c811da5d5b4bc6d497ffa98491e38
```



Crack Hashes

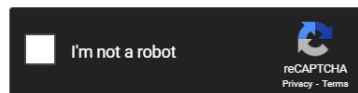
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
482c811da5d5b4bc6d497ffa98491e38	md5	password123

Color Codes: **Green:** Exact match, **Yellow:** Partial match, **Red:** Not found.

Enter up to 20 non-salted hashes, one per line:

```
b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
```



Crack Hashes

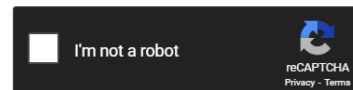
Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3	sha1	1etmein

Color Codes: **Green:** Exact match, **Yellow:** Partial match, **Red:** Not found.

Enter up to 20 non-salted hashes, one per line:

916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745	sha256	qwerty098

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

```
ameenah@AmeenahsASUS:~$ nc verbal-sleep.picoctf.net 52014
Welcome!! Looking For the Secret?

We have identified a hash: 482c811da5d5b4bc6d497ffa98491e38
Enter the password for identified hash: password123
Correct! You've cracked the MD5 hash with no secret found!

Flag is yet to be revealed!! Crack this hash: b7a875fc1ea228b9061041b7cec4bd3c52ab3ce3
Enter the password for the identified hash: letmein
Correct! You've cracked the SHA-1 hash with no secret found!

Almost there!! Crack this hash: 916e8c4f79b25028c9e467f1eb8eee6d6bbdff965f9928310ad30a8d88697745
Enter the password for the identified hash: qwerty098
Correct! You've cracked the SHA-256 hash with a secret found.
The flag is: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_7f29c9da}
```

Unhash each hash string to find the password for each stage. The flag is then revealed.

Flag: picoCTF{UseStr0nG_h@shEs_&PaSswDs!_7f29c9da}

4 PHANTOM INTRUDER

```
ameenah@AmeenahsASUS:~$ strings myNetworkTraffic.pcap
ezF0X3c0cw==0
I+znCJg=0
/9QZRtc=0
uF+2UTs=0
pDW7rkI=0
8esVOK4=0
cGljb0NURg==0
KWEh2jQ=0
BGAVCe8=0
RA+7xFw=0
YmhfNHJfZA==0
CMEx344=0
PGb6oYA=0
uSy5rvo=0
bnRfdGg0dA==0
MTA2NTM4NA==0
xIJPbWg=0
XzM0c3lfdA==0
oDis5T8=0
v2XgLLs=0
fQ==0
STneebY=
```

Strings of pcap files are analysed. The strings that end with two equal signs are decoded from base64 format. The resulting strings are rearranged to form the flag.

Decode from Base64 format

Simply enter your data then push the decode button.

```
ezF0X3c0cw==  
cGlib0NURg==  
YmhfNHJfZA==  
bnRfdGg0dA==  
MTA2NTM4NA==  
XzM0c3lfdA==  
fQ==
```

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☒ Decode each line separately (useful for when you have multiple entries).

 Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

```
{1t_w4s  
picoCTF  
bh_4r_d  
nt_th4t  
1065384  
_34sy_t  
}
```

Flag: picoCTF{1t_w4snt_th4t_34sy_tbh_4r_d1065384}

5 FLAGS ARE STEPIC (INCOMPLETE)

I know the flag is hidden in the upz.png file. It is 1.7MB large while the other images are 600KB. And it's not a country. But I can't extract useful information. Strings, ls and binwalk are not giving me any results.

```
{ name: "Tuvalu",img: "flags/tv.png" },  
{ name: "Uganda",img: "flags/ug.png" },  
{ name: "Ukraine",img: "flags/ua.png" },  
{ name: "United Arab Emirates (the)",img: "flags/ae.png" },  
{ name: "United Kingdom of Great Britain and Northern Ireland (the)",img: "flags/gb.png" },  
{ name: "United States of America (the)",img: "flags/us.png" },  
{ name: "Upanzi, Republic The",img: "flags/upz.png", style:"width: 120px!important; height: 90px!important;" },  
{ name: "Uruguay",img:"flags/uy.png" },  
{ name: "Uzbekistan",img: "flags/uz.png" },  
{ name: "Vanuatu",img: "flags/vu.png" },
```

6 EVENT VIEWING

Open up log file.

First log has installation information for Totally_Legit_Software is at 11:55:57PM. First third of flag was in base64 format.

Information	15/7/2024 11:55:55 PM	Microsoft Win...	4690	Handle ...
Information	15/7/2024 11:55:55 PM	Microsoft Win...	4658	File Sys...
Information	15/7/2024 11:55:55 PM	Microsoft Win...	4656	File Sys...
Information	15/7/2024 11:55:55 PM	Microsoft Win...	4658	File Sys...
Information	15/7/2024 11:55:55 PM	Msiinstaller	1040	None
Information	15/7/2024 11:55:57 PM	RestartManager	10000	None
Information	15/7/2024 11:55:57 PM	RestartManager	10001	None
Information	15/7/2024 11:55:57 PM	Msiinstaller	1033	None
Information	15/7/2024 11:55:57 PM	Msiinstaller	1042	None
Information	15/7/2024 11:55:57 PM	Msiinstaller	11707	None
Information	15/7/2024 11:55:58 PM	Microsoft Win...	5156	Filterin...
Information	15/7/2024 11:55:58 PM	Microsoft Win...	5156	Filterin...

Event 1033, Msiinstaller	
General	Details
Windows Installer installed the product. Product Name: Totally_Legit_Software. Product Version: 1.3.3.7. Product Language: 0. Manufacturer: cGlib0NURntFdjNudF92aTN3djNyXw==. Installation success or error status: 0.	

Decode from Base64 format

Simply enter your data then push the decode button.

cGlib0NURntFdjNudF92aTN3djNyXw==.

i For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page

UTF-8 **v** Source character set.

☒ Decode each line separately (useful for when you have multiple entries).

☐ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

picoCTF{Ev3nt_vi3wv3r_

Scrolling through the following logs. I found another part of the flag in base64 format.

Information	15/7/2024 11:56:19 PM	Microsoft Windows se...	4663	Registry
Information	15/7/2024 11:56:19 PM	Microsoft Windows se...	4657	Registry
Information	15/7/2024 11:56:19 PM	Microsoft Windows se...	4658	Registry
Information	15/7/2024 11:56:19 PM	Microsoft Windows se...	4656	Registry

Event 4657, Microsoft Windows security auditing.

General Details

A registry value was modified.

Subject:

Security ID: S-1-5-21-3576963320-1344788273-4164204335-1001

Account Name: user

Account Domain: DESKTOP-EKVR84B

Logon ID: 0x5A428

Object:

Object Name: \REGISTRY\MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run

Object Value Name: Immediate Shutdown (MXNfYV9wcjN0dHlfdXMzZnVsXw==)

Handle ID: 0x208

Operation Type: New registry value created

Log Name: Security

Source: Microsoft Windows security i

Logged: 15/7/2024 11:56:19 PM

Event ID: 4657

Task Category: Registry

Level: Information

Keywords: Audit Success

User: N/A

Computer: DESKTOP-EKVR84B

OpCode: Info

MXNfYV9wcjN0dHlfdXMzZnVsXw==

For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page

UTF-8 Source character set.

☒ Decode each line separately (useful for when you have multiple entries).

☐ Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

< DECODE > Decodes your data into the area below.

1s_a_pr3tty_us3ful_

No luck running through times right after first shutdown. Scrolled to an hour later.

Information	16/7/2024 1:02:34 AM	Microsoft Windows se...	5156	Filterin...
Information	16/7/2024 1:02:35 AM	User32	1074	None
Information	16/7/2024 1:02:36 AM	Microsoft Windows se...	4658	Remov...
Information	16/7/2024 1:02:36 AM	Microsoft Windows se...	4658	Remov...
Information	16/7/2024 1:02:37 AM	Microsoft Windows se...	4647	Logoff
Information	16/7/2024 1:02:37 AM	Winlogon	6000	None
Information	16/7/2024 1:02:37 AM	Winlogon	6000	None
Information	16/7/2024 1:02:37 AM	Winlogon	7002	(1102)
Information	16/7/2024 1:02:37 AM	Microsoft Windows se...	4700	User A...

Event 1074, User32				
General Details				
The process C:\Windows\system32\shutdown.exe (DESKTOP-EKVR84B) has initiated the shutdown of computer DESKTOP-EKVR84B on behalf of user DESKTOP-EKVR84B\user for the following reason: No title for this reason could be found Reason Code: 0x800000ff Shutdown Type: shutdown Comment: dDAwbF84MWJhM2ZlOX0=				

Decode from Base64 format

Simply enter your data then push the decode button.

dDAwbF84MWJhM2ZlOX0=

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☒ Decode each line separately (useful for when you have multiple entries).

 Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

t00l_81ba3fe9}

Flag: picoCTF{Ev3nt_vi3wv3r_1s_a_pr3tty_us3ful_t00l_81ba3fe9}

7 EVEN RSA CAN BE BROKEN???

dcode.fr/rsa-cipher



Search for a tool

★ SEARCH A TOOL ON DCODE BY KEYWORDS:
e.g. type 'random'

★ BROWSE THE FULL DCODE TOOLS' LIST

Results

- ❌ Wiener's attack: failure
- ✅ P,Q computed with N ((Self-Limited) Prime Factors Decomposition)
- ✅ D computed with P,Q,E
- ✅ Decryption using C,D,N

picoCTF{tw0_1\$_pr!m3f81fef0a}

RSA Cipher - dCode

Tag(s) : Modern Cryptography, Arithmetics

Share

dCode and more

dCode is free and its tools are a valuable help in games, maths, geocaching, puzzles and problems to solve every day!
A suggestion ? a feedback ? a bug ? an idea ? Write to

RSA CIPHER

Cryptography · Modern Cryptography · RSA Cipher

RSA DECODER

Indicate known numbers, leave remaining cells empty.

★ VALUE OF THE CIPHER MESSAGE (INTEGER) C= 7305059676226383468835184621509753288942683668429...

★ PUBLIC KEY E (USUALLY E=65537) E= 65537

★ PUBLIC KEY VALUE (INTEGER) N= 1758532906416617984554574771545394073733997904713...

★ PRIVATE KEY VALUE (INTEGER) D=

★ FACTOR 1 (PRIME NUMBER) P=

★ FACTOR 2 (PRIME NUMBER) Q=

★ INTERMEDIATE VALUE PHI (INTEGER) Φ=

★ DISPLAY ☒ PLAINTEXT AS CHARACTER STRING
☐ COMPUTED VALUES (C,D,E,N,P,Q,...)
☐ PLAINTEXT AS INTEGER NUMBER
☐ PLAINTEXT AS HEXADECIMAL FORMAT

▶ CALCULATE/DECRYPT

RSA CERTIFICATE READER

★ CERTIFICATE (STARTING WITH -----BEGIN...KEY-----)

Summary

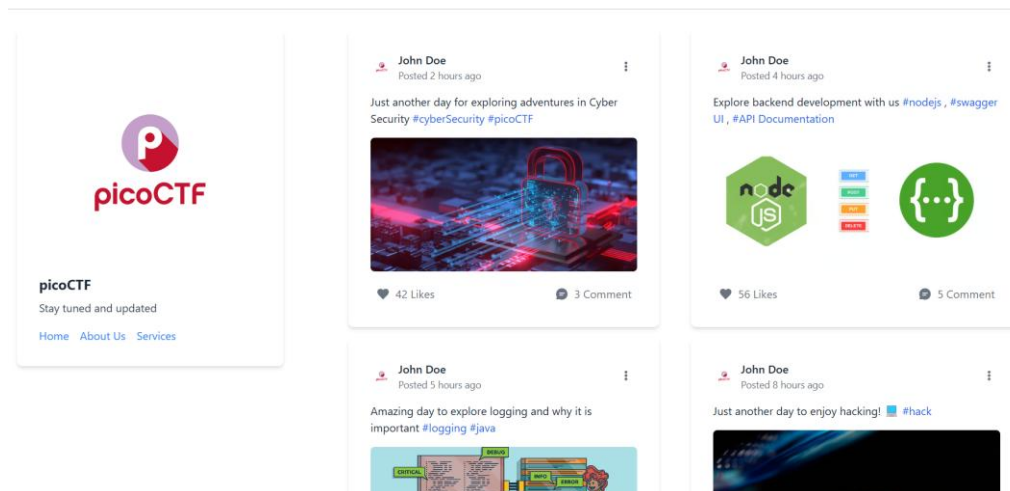
- ★ RSA Decoder
- ★ RSA Certificate Reader
- ★ Complementary Helper tools
- ★ What is the RSA cipher? (Definition)
- ★ How to encrypt using RSA cipher?
- ★ How to decrypt a RSA cipher?
- ★ How to generate RSA keys?
- ★ How to recognize RSA ciphertext?
- ★ What are possible RSA attacks?
- ★ How to decrypt RSA without the private key?
- ★ Why using the number e=65537 for RSA?
- ★ How to decrypt a number into plaintext?
- ★ What is an RSA certificate?
- ★ When was RSA invented?

Similar pages

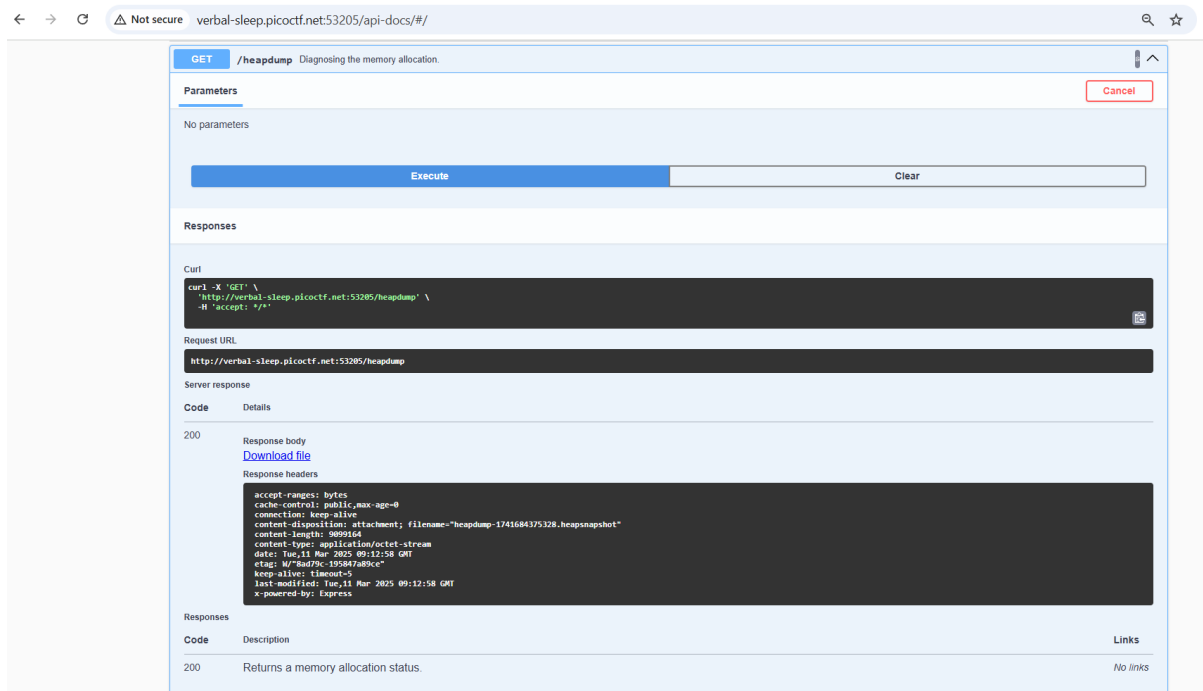
Insert e, N and ciphertext into website and decrypt to reveal flag.

Flag: picoCTF{tw0_1\$_pr!m3f81fef0a}

8 HEAD-DUMP



Click on the API documentation link. Once on the page, execute heapdump and download the file. Open it in notepad and search “pico”. This reveals the flag.



,3,2374,52674
 ,2,302,33642
 ,3,26214,265896
 ,3,137,265902
 ,3,2623,204666
 ,3,192,4290
 ,3,2374,52674
 ,2,302,33642
 ,3,26214,265908
 ,3,137,265914
 ,3,2623,204666
 ,3,

picoCTF{Pat!3nt_15_Th3_K3y_8635db4b}

192,4290
 ,3,2374,52674
 ,2,302,33642
 ,3,26214,265932
 ,3,137,265938
 ,3,2623,204666
 ,3,192,4290
 ,3,2374,52674
 ,2,302,33642

Flag: picoCTF{Pat!3nt_15_Th3_K3y_8635db4b}

9 QUANTUM SCRAMBLER

After studying the pattern of quantum scrambling for a different numbers of letters, I noticed repetition begins after 4 letters. Repetition increases in powers of two for each increase in letter.

Mapped the word “pico” to its hexadecimal characters to start off. Then replaced remaining hexadecimal values. Sample shown below.

```

[['p', 'i'], ['c', []], 'o'],
[C, [['p', 'i'], 'T'],
[F, [['p', 'i'], ['c', []], 'o']], 'f',
[p, [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T']], 'y'],
[t, [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], 'h']
, ['o'], [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p', 'i'], ['c', []], 'o'], ['C',
[['p', 'i'], 'T']], 'y']], 'n'],

['s', [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p', 'i'], ['c', []], 'o'], ['C',
[['p', 'i'], 'T']], 'y'], ['t', [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], 'h'], ['o',
[['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p', 'i'], ['c', []], 'o'], ['C', [['p',
'i'], 'T']], 'y']], 'n']], 'l'],

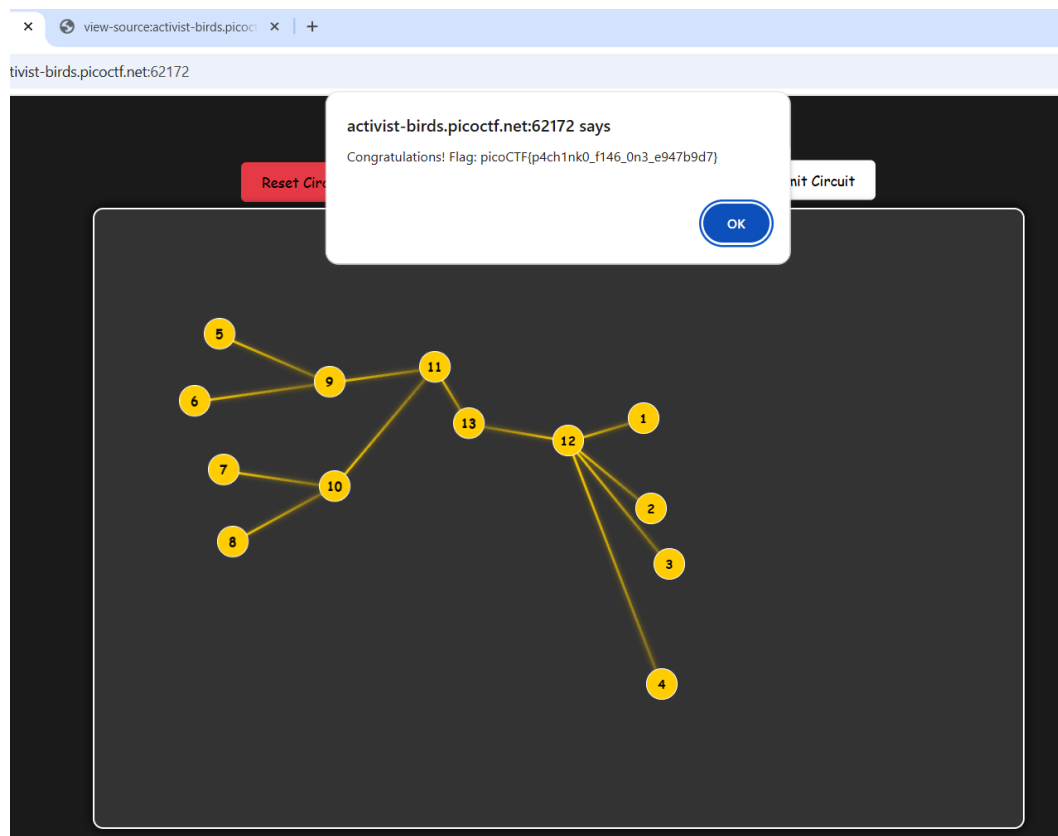
[w, [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p', 'i'], ['c', []], 'o'], ['C',
[['p', 'i'], 'T']], 'y'], ['t', [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], 'h'], ['o',
[['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p', 'i'], ['c', []], 'o'], ['C', [['p',
'i'], 'T']], 'y']], 'n'], ['_', [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []], 'o']], '{'], ['p', [['p',
'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T']], 'y'], ['t', [['p', 'i'], ['c', []], 'o'], ['C', [['p', 'i'], 'T'], ['F', [['p', 'i'], ['c', []],
'o']], '{'], 'h']], 'i']], 'e'],

```

Flag: picoCTF{python_is_weirdd91a5340}

10 PACHINKO

To understand the game, I looked at the code and did some research about Pachinko. In order to win the game, the inputs node values need to be flipped at the output. This needs to be achieved using the NAND gates which can be added by pressing the “Add intermediate Node” button. I needed to try several circuits until I got the flag.



Flag: picoCTF{p4ch1nk0_f146_0n3_e947b9d7}

11 TAP INTO HASH

The steps used to encrypt the block chain are identified from the `block_chain.py` file. A random hex chain is first generated and converted to bytes. This is used as the key for the final encryption step. The block chain is then generated and a token is inserted into the middle. In this case the token is the picoCTF flag. Then it is padded to multiple blocks, each sized 16-bytes. Finally, each 16-byte block is XORed with the SHQ-256 hash of the key.

In order to decrypt the given block chain, the encryption steps need to be reversed. A python program is used to do that.

```

C: > Users > ameen > Desktop > Ameenah > Self study > picoCTF 2025 > decrypt_blockchain.py > ...
1  import hashlib
2
3  # Given key and encrypted blockchain
4  key = b'\x8b\x9a\x06\xfe\xb3\xfb\x93\xdb\xa8yT\xfe\x15\x87a\xfb\xdf\x00\x8d\xee\xab\xdb\t^\x04(\x81\x9e\xfb'
5  encrypted_blockchain = b'Z5Wo\x09\xbd\xfb\xed\xeb\xcb\xcc\xfb\x02\x0b1\x09\x09\xfb\x08\x08n\x91q\xca\xad=\x01fRo\xbe\xba
6
7  # Step 1: XOR Decryption
8  def xor_bytes(a, b):
9      return bytes(x ^ y for x, y in zip(a, b))
10
11  def decrypt(ciphertext, key):
12      key_hash = hashlib.sha256(key).digest()
13      plaintext = b''
14
15      for i in range(0, len(ciphertext), 16):
16          block = ciphertext[i:i + 16]
17          plaintext += xor_bytes(block, key_hash)
18
19      return plaintext
20
21  # Step 2: Remove Padding
22  def unpad(data):
23      padding_length = data[-1]
24      return data[:-padding_length]
25
26  # Step 3: Decrypt and Unpad
27  padded_plaintext = decrypt(encrypted_blockchain, key)
28  plaintext = unpad(padded_plaintext)
29
30  # Step 4: Extract the Original Blockchain String
31  # The flag is inserted in the middle of the plaintext
32  print("Decrypted Blockchain String:", plaintext.decode('utf-8'))

```

The flag is found in the middle of the decrypted block chain.

```

23  padding_length = data[-1]
24  return data[:-padding_length]
25
26  # Step 3: Decrypt and Unpad
27  padded_plaintext = decrypt(encrypted_blockchain, key)
28  plaintext = unpad(padded_plaintext)
29
30  # Step 4: Extract the Original Blockchain String
31  # The flag is inserted in the middle of the plaintext
32  print("Decrypted Blockchain String:", plaintext.decode('utf-8'))

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS JUPYTER

```

PS C:\Users\ameen\Desktop\Ameenah\Self study\Python\Drug_discovery> & C:/Python312/python.exe "c:/Users/ameen/Desktop/Ameenah/Self study/picoCTF 2025/decrypt_blockchain.py"
● Decrypted Blockchain String: 85dbbeaaffc2894128ab1edae59f6d7cfab5b7995c8c8ee12c8e483cbabb80-008f3cf132d9c72b87ad6ae82c3cccd8685dc34ad5e3992beb3d5af90667773e-001a7039a41bb27460c41f0b044fedbbpicoCTF{block_3SRhViRbT1qcX_XUjM0r49cH_qCzmJZzBK_60647fbb}ce014e7d81f1d569d360c9706a1c98ee-0066a2261eabb27ed179a377bfe4a285d2d8acff133bee199e55a116a5f5a533-00fda7226a6f35003bb8f4c0c6ab004221a892aab38608fbed9f5adc2690b253
○ PS C:\Users\ameen\Desktop\Ameenah\Self study\Python\Drug_discovery>

```

Flag: picoCTF{block_3SRhViRbT1qcX_XUjM0r49cH_qCzmJZzBK_60647fbb}